Excel-Based Product Inventory Data Cleaning & Analysis

Submitted By:

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Course

Program in AI-Driven Data Analytics

Project Objective

Clean, transform, and analyse a raw product inventory dataset using Excel to extract actionable business insights. This project demonstrates core data cleaning, formatting, and basic analysis techniques using formulas in Excel.

Dataset Description

The dataset contains 20+ records of global product inventory with details including manufacturing date, country of origin, product brand, price, stock quantity, and category. It simulates real-world e-commerce/inventory data with formatting inconsistencies and missing values.

Column Description

Column Name	**Description**
Manufacturing Date	Date when the product was manufactured. Standardized to dd-mm-yyyy format.
Country Code	ISO country code indicating product origin: US, UK, IN, AU, DE, CA, ES, CN, IT.
Product Brand	Combined product name and brand. Split and merged during cleaning for better structure.
Price ($)	Unit price of the product. Handled missing values and standardized currency format.
Quantity	Stock quantity available for each product.
Category	Product category: Electronics, Fashion, Kitchen, Outdoor, Accessories, Unknown.

Data Cleaning & Transformation Steps Performed:

1. Handled Missing Data:

- Imputed missing Price values with 0 and flagged for review
- Filled blank Category entries with Unknown to maintain data integrity

2. Standardized Text & Formatting:

- Cleaned text fields using TRIM, CLEAN, and PROPER functions to remove extra spaces and standardize capitalization
- Corrected spelling errors and typos in the Category column for consistent grouping
- Applied currency formatting to Price and DD-MM-YYYY formatting to Manufacturing Date.

3. Restructured Key Fields:

- Split Product ID into separate Manufacturing Date and Country Code columns for better analysis
- Merged Brand Name and Product Name into a single Product Brand column for readability

4. Improved Data Quality:

- Removed duplicate records using Excel's Remove Duplicates tool
- Applied conditional formatting to Price and Category columns to quickly identify outliers and categories.

Outcome:

Transformed raw inventory data into a clean, analysis-ready dataset suitable for pivot tables and visualizations.

Visualizations & Key Insights

To make the data easier to interpret, I used Excel's conditional formatting and visual cues:

1. **Price Distribution:**
   - Applied conditional formatting to the Price column to quickly identify high and low-priced products. This revealed a clear price range across categories.
2. **Category Breakdown:**
   - Highlighted the Electronics category in green to visually separate it from others.
3. **Key Insights Identified:**

   Price Trend: Electronics products consistently showed the highest unit prices compared to Fashion, Kitchen, and Accessories.

   Category Mix: Electronics made up the largest share of products in the dataset, indicating it's the dominant category in this inventory.

Outcome:

The visualizations made it easy to spot pricing patterns and category concentration at a glance, demonstrating how basic Excel formatting can drive quick, actionable insights.

	Manufacturing Dat	Country Cod	Product Brand	Price ($	Quantit	Category			
1								Tasks:	
2	28-01-2025	US	Dell Laptop	$1,000	30	Electronics			
3	15-02-2025	US	Nike Sneakers	$80	15	Fashion		1. Handling Missing Values:	6. Conditional Formatting:
4	03-03-2025	US	Keurig Coffee Maker	$130	40	Kitchen			
5	11-04-2025	US	Samsung Smartphone	$900	25	Electronics		column.	Conditional Formatting " Price"
6	22-05-2025	US	North Face Backpack	$70	20	Unknown	Ans:	1000	Price column.
7	07-06-2025	UK	Sony Headphones	0	45	Electronics			
8	19-07-2025	UK	Adidas T-Shirt	$30	5	Fashion		categories,	Conditional Formatting "Category"
9	23-08-2025	UK	Ninja Blender	$90	35	Kitchen	Ans:	Electronics	formatting rule.
10	05-09-2025	UK	Apple Tablet	$500	50	Electronics			
11	14-10-2025	UK	Timberland Hiking Boots	$130	10	Outdoor		2. Correcting Inconsistent Data:	
12	17-06-2025	IN	HP Laptop	$950	25	Electronics			
13	25-11-2025	AU	Adidas Sneakers	$90	40	Unknown		present in the "Product Name" column.	in the "Category" column.
14	08-12-2025	DE	Nespresso Coffee Maker	$120	35	Unknown	Ans:	PROPER(TRIM(CLEAN()))	
15	18-02-2025	CA	Fitbit Smartwatch	$150	15	Electronics			misspellings.
16	16-04-2025	ES	Bose Headphones	$250	20	Electronics		b)Identify any typos present in the "Category" column.	
17	21-08-2025	CA	Samsonite Laptop Bag	$50	35	Accessories	Ans:	ctrl+h-Electroni-Electronics	
18	20-08-2025	CN	Huawei Smartwatch	$160	15	Electronics			
19	27-01-2025	IT	Asus Laptop	$980	10	Electronics		3. Removing Duplicates:	
20	01-03-2025	UK	Oakley Sunglasses	$150	15	Fashion			
21	14-08-2025	US	Coleman Camping Tent	0	10	Outdoor	Ans:	using Remove Duplicates option.	
22	14-05-2025	RU	Nikon Camera	$700	50	Electronics			
23	09-01-2025	CA	Panasonic Microwave	$80	20	Kitchen		4. Splitting and Merging Data:	
24	19-07-2025	BR	Xiaomi Fitness Tracker	$150	30	Unknown			
25	21-08-2025	CA	Samsonite Laptop Bag	$50	35	Accessories	Ans:	unnecessary characters.	
26	29-09-2025	CA	Google Smartphone	$800	45	Electronics			
27	03-06-2025	CA	Ray-Ban Sunglasses	0	25	Fashion			
28	11-07-2025	CA	Vitamix Blender	$400	40	Kitchen	Ans:	into a new column called Product Brand.	
29	16-04-2025	ES	Bose Headphones	$250	20	Electronics			
30	07-03-2025	CA	Zara Dress	$60	30	Fashion		5. Number Formatting:	
31	13-04-2025	CA	Hamilton Toaster	$40	10	Kitchen		a)Number Formatting "Price"	
32	24-05-2025	CA	Garmin Fitness Tracker	$130	5	Electronics	Ans:	currency format.	
33	02-12-2025	CA	Levi's Jeans	$50	50	Fashion			
34	17-06-2025	IN	HP Laptop	$950	25	Electronics		Date	
35	09-07-2025	FR	Casio Watch	$100	20	Accessories	Ans:	format.	
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Analysis & Insights

Descriptive Analysis:

Summary statistics revealed key characteristics of the inventory dataset:

- **Price Range**: ₹0 – ₹1000, with ₹0 indicating missing or placeholder values that require review
- **Category Distribution**: Electronics was the most common category, followed by Fashion and Kitchen
- **Stock Levels:** Maximum quantity recorded was 50 units

Diagnostic Analysis:

Identified data quality issues impacting reliability:

- Missing values and typos in Category/Price columns introduced inconsistencies
- Duplicate records were present, which could inflate counts and distort analysis

Predictive Analysis:

Based on cleaned data patterns:

- Electronics are likely to remain the highest-priced category due to current pricing trends

  Products with higher stock quantities may correlate with higher demand or faster-moving inventory.

Prescriptive Analysis:

To maintain data quality going forward:

- Enforce standardized category naming to avoid typos and misclassification
- Validate Price and Quantity fields during data entry to reduce missing values
- Schedule regular de-duplication checks to ensure analysis accuracy

Conclusion

The raw inventory dataset was successfully cleaned, standardized, and transformed using Excel. After addressing missing values, duplicates, and formatting issues, the data is now structured and analysis-ready for reporting, visualization, and deeper business insights.